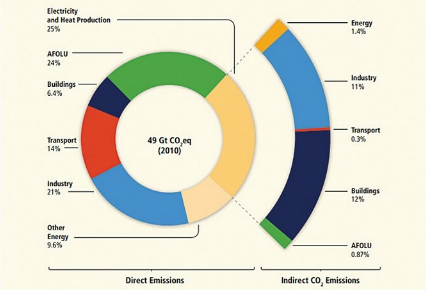

Figure 2: The global greenhouse gas emission in 2010 (*Image Source: www.ipcc.ch*)

greenhouse gas emissions. This would also mean limiting our energy consumption, including electricity, coal, and oil usages.

Table 1 shows the United States' greenhouse gas emissions by a few sectors in 2017. As per the data from the Intergovernmental Panel on Climate Change (IPCC), electricity and heat production accounted for 25 per cent of global greenhouse gas emissions in 2010 (Figure 2). These include the burning of coal, natural gas, and oil. On the other hand, agriculture, forestry, and other land use accounted for 24 per cent of greenhouse gas emissions in 2010.

It is of utmost importance to focus on these major areas for new technology development in order to control climate change. This article covers how new technologies like artificial intelligence (AI) and machine learning (ML) can be used to study the environment and control climate change.

The role of AI in fighting climate change

AI can play an important role in fighting climate change. Last year, a group of the world's most prominent AI experts published a detailed paper titled 'Tackling Climate Change with Machine Learning.' It covers how AI and ML can help accelerate various strategies to fight against climate change.

Areas	Greenhouse emission percentage
Transportation	29.1%
Electricity generation	27.7%
Industry	22.4%
Commercial	20%
Agriculture	9.1%
Residential	5.2%

Table 1: Some areas of areas of greenhouse emissions in the US in 2017

There are different approaches to using AI to study the environment and control climate change. Approaches that are rule-based and learning-based can be used. Rule-based AI helps scientists compile CO_2 emission data. Learning-based AI is more advanced than rule-based AI because it can interact with problems, diagnose, and recommend solutions. That is, learning-based AI can not only compile CO_2 emission numbers but also study the causes and then recommend the best solutions. These approaches have been used in many uses cases and applications.

Use cases of AI in fighting climate change

Some of the use cases of AI in controlling climate change and related applications are given below.

Global Forest Watch: It is conceived as an open source Web application to identify deforestation in real time using satellite and AI. It provides deforestation details with various features such as country-wise or particular geographical area, forest fire alert with time/date, information related to climate change, etc. For example, a close look at the forest fire alert in the US in September captured through the Global Forest Watch website *https://www.globalforestwatch.org* is shown in Figure 3.

SilviaTerra: This startup company in the US provides next-generation, highly accurate forest inventory data. It is powered by Microsoft technology. It uses AI and satellite imaging to predict sizes, species, and health of forest trees. The health of forests is important to help fight climate change.

Sidewalk Labs: This company is harnessing digital technologies to its urban traffic problems. One of its projects focuses on how traffic flows in a city and checks the hotspot of congestion. The goal is to reduce air pollution and improve efficient transportation in a city.

CycleGAN: The Cycle Generative Adversarial Network or CycleGAN is basically a technique to transfer characteristics of one image to another. A student at the University of Montreal first invented Generative Adversarial Networks (GANs) in 2014. CycleGAN is an approach to train a deep convolutional neural network for image-to-image translation tasks. It uses AI to train itself to produce images that portray geographical locations before and after extreme weather conditions. The final images produced could help scientists predict the impact of certain climate changes, thereby helping humans to take appropriate actions.